NFH-Kuza-Google Pilot

Master Requirements Specification

Single-Window Rural Service Delivery Marketplace

Kuza — Platform | Google — Cloud & AI | Beckn — Open Network

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1. Executive Summary

NFH-Kuza-Google is a pilot project to build a **single-window rural service delivery marketplace**. It connects rural households with local service providers through **WhatsApp** (farmer side) and a **web app** (provider side), using the **Beckn open network protocol** for interoperability.

> **PARTNERS**
>
> **Kuza** — Platform technology, farmer & provider networks, domain expertise.
>
> **Google** — Three contributions: (1) Cloud infrastructure (GCP) for hosting the Kuza stack, (2) Content APIs serving curated domain content (agri, health, market, finance) through the chatbot, (3) AI interfaces (location normalisation, content tailoring & generation).
>
> **Beckn** — Open network protocol, DeDi Registry, ONIX adapters, Catalog & Discovery Services.

Key Characteristics

Characteristic	Description
Rural-focused	Designed for cash-and-carry transactions. No e-commerce features (shopping cart, shipping, online payments).
WhatsApp-first	The farmer's entire experience is through WhatsApp — no app installation required.
Provider hybrid	Providers receive WhatsApp notifications and manage operations through a browser-based web app (no app installation).
Cross-domain	A single provider can offer services across all four categories simultaneously.
Single window	Everything a rural household needs — livelihood services, household services, market access, advisory content — through one interface.
Location-based	Providers matched by registered service areas (County + Sub-County).
Kenya pilot	47 Counties / Sub-Counties, KES currency, M-Pesa mobile money.

2. Actors

2.1 Farmers / Rural Households (Seekers)

- **Primary interface:** WhatsApp
- **Actions:** Discover services, select providers, confirm orders, confirm service delivery, report payment method, provide feedback
- **Profile captured once:** name, phone (auto from WhatsApp), village/ward/county, GPS (optional), primary enterprise (optional), preferred language (optional)

2.2 Service Providers / Agripreneurs / Aggregators (Providers)

- **Primary interface:** WhatsApp (notifications + quick actions) + Kuza Web App (detailed management)
- **Actions:** Receive order notifications on WhatsApp with action buttons; manage catalog, accept/reject orders, update status, report payment, provide feedback via web app
- **Profile:** name, phone, service area (counties/sub-counties), services offered, price list, capacity, availability, provider type
- A **single provider can serve across all four categories**
- Agripreneurs already exist in Kuza Platform; other providers need onboarding

3. Service Categories & Taxonomy

Domain code: `local-services` (Beckn v2)

Four top-level categories with dynamic sub-categories managed as master data:

Category	Example Sub-Categories	Example Services
Livelihood-Services	Livestock Services, Crop Services, Trades & Skills	Vaccination, brooding, deworming, feed supply, ploughing, spraying, tailoring, welding, carpentry
Household-Services	Education, Health, Insurance, Consumer Goods	School fees, tutoring, books, hospital checkup, pharmacy, crop/health insurance, household items
Market-Produce	Farm Produce, Livestock Products, Handicrafts & Goods	Vegetables, grains, fruits, eggs, milk, meat, live birds, handmade goods
Advisory-Content	Agricultural Advisory, Financial Literacy, Health & Nutrition, Business Skills	Poultry care, crop management, budgeting, nutrition guides

DYNAMIC TAXONOMY

Sub-categories are **not** hardcoded. They are managed as master data in Kuza Platform and pulled dynamically. New sub-categories can be added without schema changes.

4. Detailed Flow Specifications

4.1 Livelihood Services Flow (Full Lifecycle)

Example: Poultry Vaccination in Nakuru County

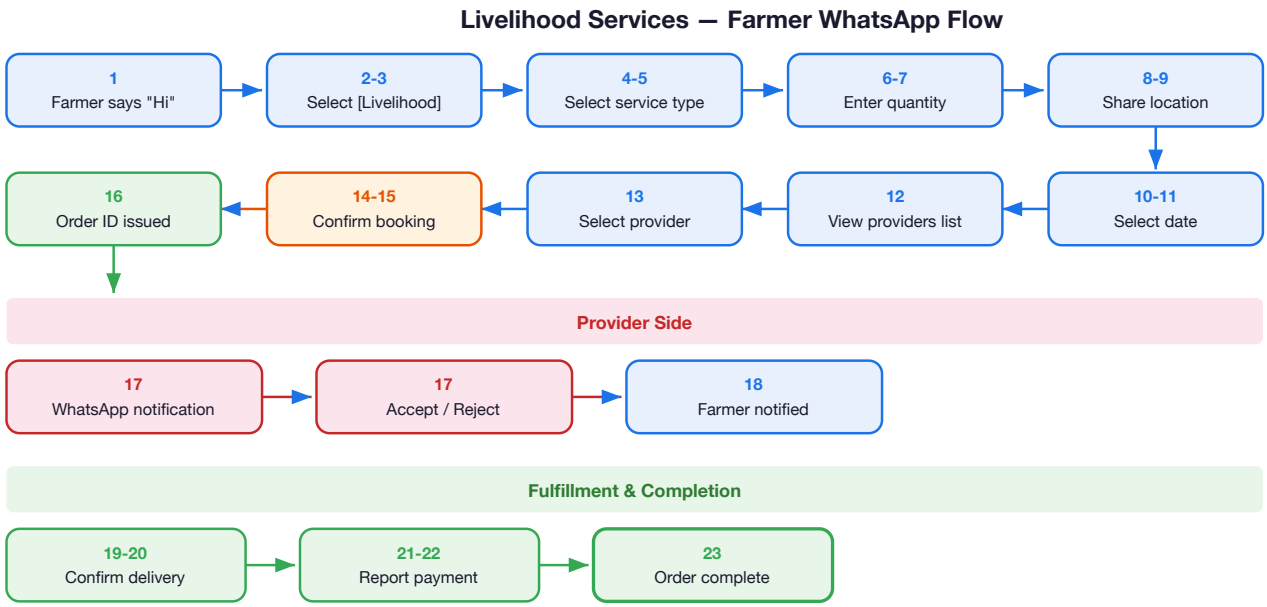

Figure 4.1 — Livelihood Services: End-to-end flow from farmer request to completion

Farmer WhatsApp Flow (Step by Step)

1. Farmer sends **"Hi"**
2. System: "*Welcome. What do you need help with?*" — 4 buttons: Livelihood, Household, Market, Advisory
3. Farmer selects **[Livelihood]**
4. System: "*What service do you need?*" — Numbered list: 1. Vaccination 2. Inputs 3. Brooding 4. Incubation (pulled from master data)

5. Farmer selects "1" (Vaccination)
6. System: "*How many birds do you need vaccinated?*"
7. Farmer enters number (e.g., "200")
8. System: "*Please share your location*" — farmer shares village/ward/county or live GPS location
9. Google AI normalises location. System confirms: "*Nakuru County, Naivasha Sub-County. Correct?*" [Yes] [No]
10. System: "*When do you need the service?*" — buttons: Today, Tomorrow, Choose date
11. Farmer selects preferred date
12. System: "*Available service providers near you:*" — Numbered list showing provider name, price, rating
    `1. John Kariuki - KES 30/bird - 4.5 stars`
    `2. Peter Omondi - KES 25/bird - 4.2 stars`
13. Farmer selects provider (e.g., "1")
14. System shows booking summary: Service, Date, Provider, Total Cost (200 birds x KES 30 = KES 6,000) — [Confirm] [Cancel]
15. Farmer taps **[Confirm]**
16. System: "*Your request has been sent to the provider. Order ID: ORD-20260412-001*"
17. Provider receives WhatsApp notification with Accept/Reject buttons
18. System updates farmer: "*Your request has been accepted by John Kariuki. Visit on 12 Apr 2026. Contact: +254 712 345 678*"
19. After service, System asks: "*Was the vaccination completed?*" [Yes] [No]
20. Farmer taps **[Yes]**
21. System asks: "*How did you pay?*" — buttons: [Cash] [M-Pesa]
22. Farmer selects payment method
23. System: "*Thank you. Your request is completed.*"

Provider Flow for Same Order

1. Provider receives WhatsApp: "*New order received! Vaccination - 200 birds. Farmer: Mary Wanjiku, Naivasha, Nakuru. Date: 12 Apr 2026. Amount: KES 6,000*" — [Accept] [Reject] [View Details]
2. Provider taps **[Accept]** → order status changes to `PROVIDER_ACCEPTED` → farmer notified
3. Provider taps **[View Details]** → opens Kuza Web App showing full order

4. After completing service, provider opens web app → marks complete → enters: actual service date, payment received (yes/no), amount, payment method, notes ("Vaccinated 200 birds, all healthy"), follow-up required (yes/no)
5. System triggers farmer confirmation flow (steps 19–23 above)

Beckn API Mapping

Flow Step	Beckn Action	Direction	Payload Summary
Steps 3–11 (category + service + location + date)	`discover`	BAP → Network	category, subCategory, quantityRequired, farmerLocation, scheduledDate
Step 12 (provider list)	`on_discover`	Network → BAP	Catalog with providers, items (LivelihoodItemAttributes), offers filtered by service area
Step 13 (select provider)	`select`	BAP → BPP	Selected provider ID + item ID + offer ID
Quote confirmation	`on_select`	BPP → BAP	Price quote, cancellation policy, provider details
Steps 14–15 (confirm booking)	`init` + `confirm`	BAP → BPP	Farmer contact, scheduled date, quantity, confirmation
Step 16 (order created)	`on_confirm`	BPP → BAP	Order ID, provider details, status=ORDER_PLACED
Steps 17–18 (provider accepts)	`status` update	BPP → BAP	status=PROVIDER_ACCEPTED, confirmedVisitTime
Steps 19–22 (completion + payment)	`status` update	Both	status=COMPLETED → FARMER_CONFIRMED, payment details
Feedback	`rate` / `on_rate`	Both	Feedback objects

4.2 Household Services Flow (Full Lifecycle)

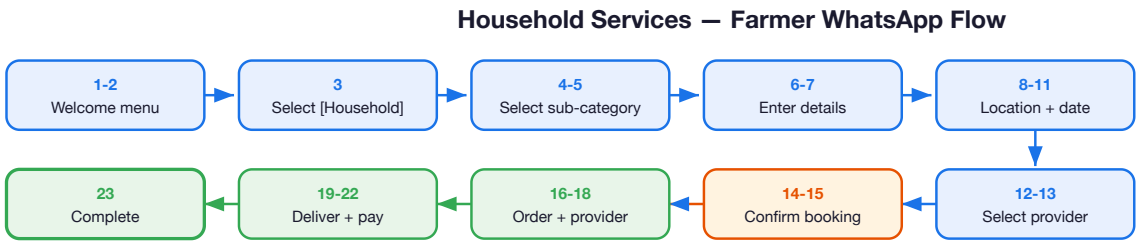

Figure 4.2 — Household Services: Structurally identical to Livelihood with domain-specific details

Structurally identical to Livelihood. Key differences:

- **Sub-categories:** Education, Health, Insurance, Consumer Goods
- **Step 6 equivalent:** Quantity/details question varies by service type (e.g., "Which school?" for fees, "How many items?" for products)
- `serviceMode` may be `FACILITY_VISIT` (hospital), `PRODUCT_DELIVERY` (items), `SERVICE_DELIVERY` (insurance agent)
- `beneficiaryType` can indicate CHILD, ELDERLY, etc.
- Some services may require documents (school admission letter, hospital card)

Example: School Fees Payment in Kiambu

1. Steps 1–5: Same pattern — select Household → Education → School Fees Payment
2. Step 6: "*Which school and class?*" (free text)
3. Step 7: "*School fee amount?*"
4. Steps 8–11: Location + date (same pattern)
5. Steps 12–23: Provider selection → confirmation → fulfilment → payment (same pattern)

4.3 Market Produce Flow (Full Lifecycle — Reversed Economics)

> **KEY DIFFERENCE**

> In this flow the farmer is the **seller** and the provider is the **buyer/aggregator**. Money flows from provider to farmer.

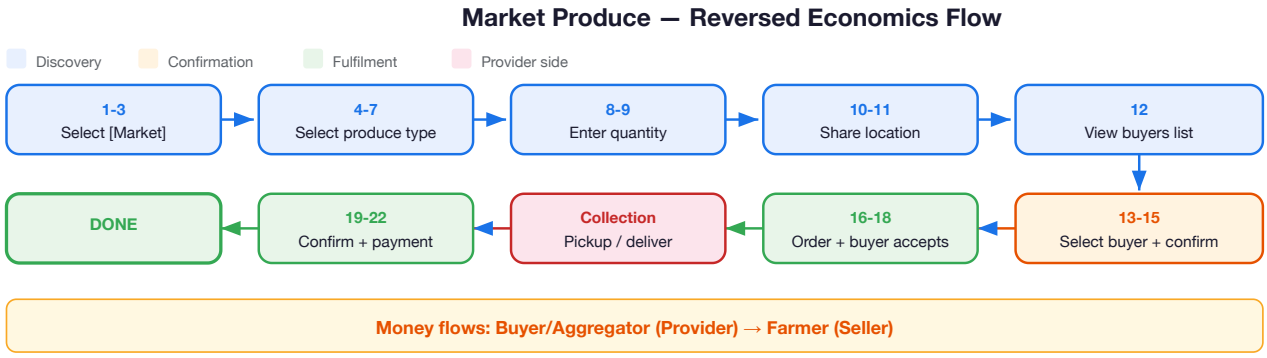

Figure 4.3 — Market Produce: Farmer is the seller, provider is the buyer. Money direction is reversed.

Example: Joseph in Kiambu has 20 trays of eggs to sell

1. Steps 1–2: Same welcome
2. Farmer selects **[Market]**
3. "*What would you like to sell?*" — 1. Farm Produce 2. Livestock Products 3. Handicrafts & Goods
4. Farmer selects "2" (Livestock Products)
5. "*What livestock product?*" — 1. Eggs 2. Milk 3. Live Birds 4. Meat
6. Farmer selects "1" (Eggs)
7. "*How many trays of eggs do you have?*"
8. Farmer enters "20"
9. Steps 10–11: Location (same pattern)
10. "*Here are buyers near you:*" — `1. Mama Eggs Traders - KES 380/tray - Farm gate pickup - 4.3 stars`
11. Farmer selects buyer
12. Summary: "*Selling: Eggs - 20 trays. Buyer: Mama Eggs Traders. Price: KES 380/tray (Total: KES 7,600). Collection: Farm gate pickup*" — [Confirm] [Cancel]

13. Steps 15–23: Same confirmation → provider notification → fulfilment → payment confirmation pattern

Key Schema Differences

- `discover` intent includes `produceCategory`, `quantityAvailable`
- `on_discover` returns aggregators with **buying prices**, `collectionMode`, `minimumQuantity`
- In `OrderAttributes`, `amountPaid` represents what aggregator pays farmer (reversed direction)
- `collectionMode` drives fulfilment: `FARM_GATE_PICKUP` | `COLLECTION_POINT` | `FARMER_DELIVERS`

4.4 Advisory Content Flow (Discover-Only)

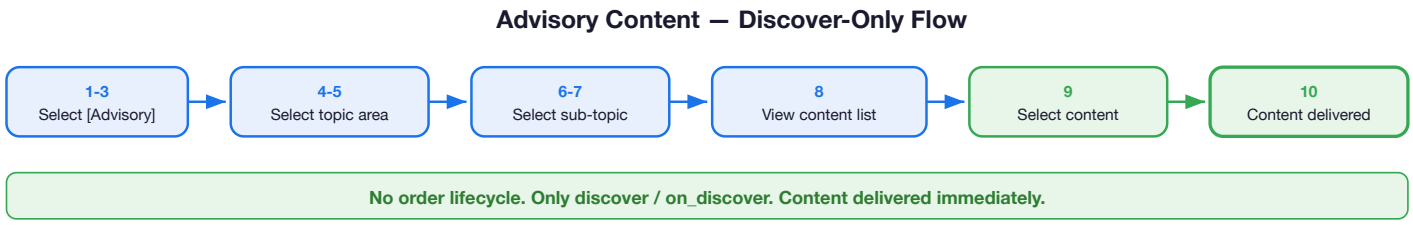

Figure 4.4 — Advisory Content: Discover-only, no transaction lifecycle

Example: Ruth in Bungoma wants to learn about poultry disease prevention

1. Steps 1–2: Same welcome
2. Farmer selects **[Advisory]**
3. "*What topic are you interested in?*" — 1. Agricultural Advisory 2. Financial Literacy 3. Health & Nutrition 4. Business Skills
4. Farmer selects "1"
5. "*Agricultural advisory topics:*" — 1. Poultry Care 2. Crop Management 3. Soil Health 4. Market Readiness
6. Farmer selects "1" (Poultry Care)
7. System sends request through Beckn (BAP → BPP). BPP calls **Google Content APIs** for the chosen domain (agri / health / market / finance) with category and sub-category to retrieve curated content.

8. BPP optionally calls **Google AI Interfaces** to personalise / summarise / translate the content based on farmer context (location, enterprise, language).
9. BPP composes the final advisory and returns it through Beckn (BPP → BAP) in the `on_discover` callback.
10. System delivers the advisory content to farmer on WhatsApp. "*Was this helpful?*" [Yes] [No]

Beckn mapping: `discover` / `on_discover`. The BPP calls Google AI during `on_discover` processing to generate advisory content. The AI-generated response is returned in the `on_discover` callback. Advisory is free — no `select` / `init` / `confirm` / `status`.

> **Google in Advisory flow:** Unlike the other three flows where content comes from provider catalogs in Kuza Platform, Advisory content is sourced from **Google Content APIs** (curated knowledge bases per domain — agri, health, market, finance) and optionally tailored through **Google AI Interfaces** (personalisation, summarisation, translation based on farmer context). The BPP calls both as part of processing the `discover` request and returns the composed advisory in `on_discover`.

5. Data Model

5.1 Farmer Profile (One-time Fields)

Field	Type	Source	Required
`farmer_name`	string	WhatsApp registration	Yes
`phone_number`	string	Auto from WhatsApp (`wa_id`)	Yes
`village`	string	Captured once	No
`ward`	string	Captured once	No
`county`	string	Captured once	Yes
`sub_county`	string	Captured / derived	Yes
`gps_latitude`	decimal	Optional live location	No
`gps_longitude`	decimal	Optional live location	No
`primary_enterprise`	string	Optional	No
`preferred_language`	string	Optional (ISO 639-1)	No

5.2 Provider Profile (One-time Fields)

Field	Type	Source	Required
`provider_name`	string	Registration / onboarding	Yes
`phone_number`	string	Auto from WhatsApp	Yes
`provider_type`	string	Registration (vet, agrovet, aggregator, trainer, etc.)	Yes
`services_offered`	relation	Links to catalog items	Yes
`price_list`	relation	Links to offers per service	Yes

Field	Type	Source	Required
`service_areas`	relation	Counties + Sub-Counties served	Yes
`capacity`	object	`{value, unit, period}` per service	No
`availability`	object	`{availableDays, timeSlots}`	No

5.3 Transaction Fields — Farmer Side (per request)

Field	Type	Captured When
`service_needed`	string	Steps 4–5
`quantity / details`	varies	Steps 6–7 (e.g., number of birds)
`preferred_date`	date	Steps 10–11
`selected_provider`	relation	Step 13
`booking_confirmed`	boolean	Step 15
`order_id`	string	Step 16 (system generated)
`order_status`	enum	Throughout lifecycle
`service_confirmed`	boolean	Step 19
`payment_method`	enum (`CASH` | `MPESA`)	Step 21
`payment_status`	enum	Step 22

5.4 Transaction Fields — Provider Side (per request)

Field	Type	Captured When
`request_id`	string	On notification
`service_requested`	string	From order
`farmer_location`	object	From order
`quantity`	varies	From order

Field	Type	Captured When
`requested_date`	date	From order
`acceptance_status`	enum (`ACCEPTED` | `REJECTED`)	Provider action
`confirmed_visit_time`	datetime	Provider confirms
`fulfillment_status`	enum (`COMPLETED` | `NOT_COMPLETED`)	After service
`payment_received`	boolean	Provider reports
`amount_received`	number	Provider reports
`notes_from_visit`	string	Provider reports
`follow_up_required`	boolean	Provider flags

6. Service Status Lifecycle

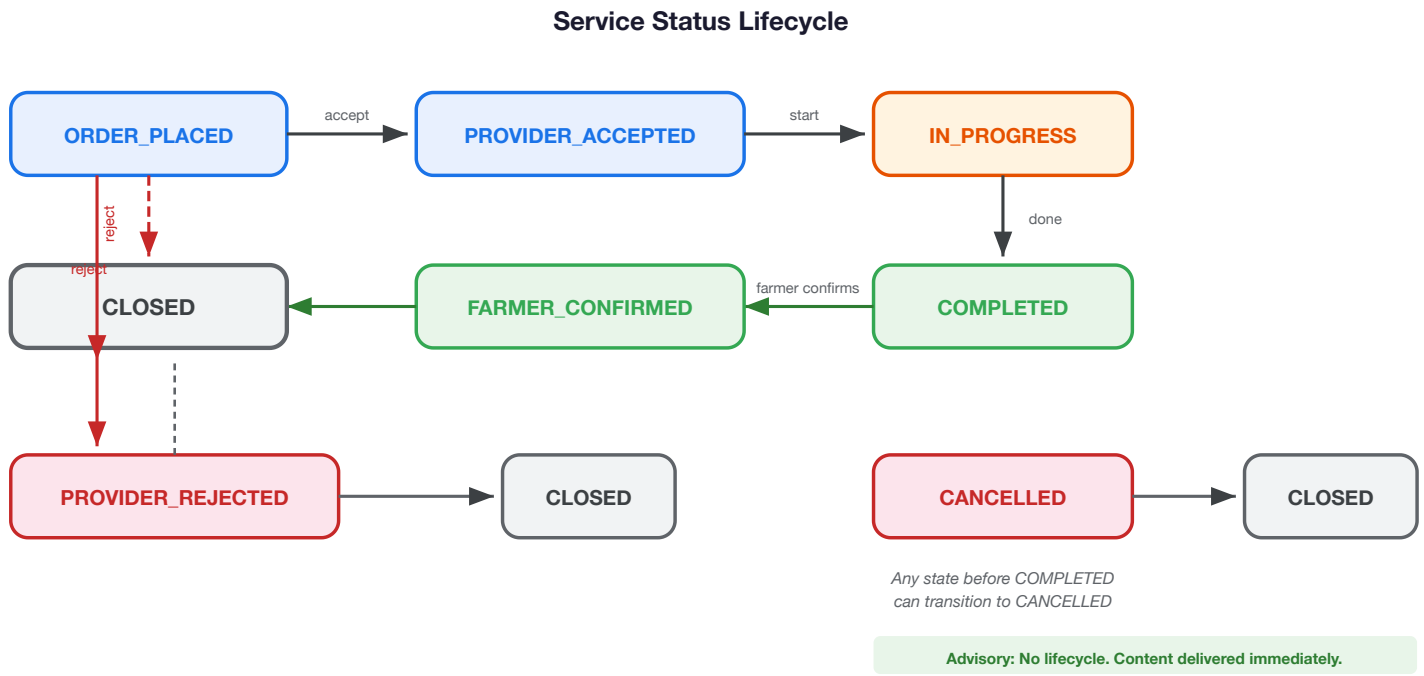

Figure 6.1 — Service status lifecycle state machine. Green path = happy path. Red = rejection/cancellation.

State Definitions

State	Description	Triggered By
`ORDER_PLACED`	Order created after farmer confirms	Farmer taps [Confirm]
`PROVIDER_ACCEPTED`	Provider accepts the order	Provider taps [Accept]
`PROVIDER_REJECTED`	Provider rejects the order	Provider taps [Reject]
`IN_PROGRESS`	Provider has started the service	Provider updates via web app
`COMPLETED`	Provider marks service as done, reports payment	Provider updates via web app
`FARMER_CONFIRMED`	Farmer confirms service was delivered and reports payment method	Farmer taps [Yes] via WhatsApp
`CANCELLED`	Either party cancels before COMPLETED	Farmer or provider action

State	Description	Triggered By
`CLOSED`	Terminal state	System (after FARMER_CONFIRMED, PROVIDER_REJECTED, or CANCELLED)

Transition Rules

```
ORDER_PLACED  ──→  PROVIDER_ACCEPTED  ──→  IN_PROGRESS  ──→  COMPLETED  ──→  FARMER_C
ONFIRMED  ──→  CLOSED
     │
     └──→  PROVIDER_REJECTED  ──→  CLOSED

Any state before COMPLETED  ──→  CANCELLED  ──→  CLOSED
```

7. WhatsApp Message Formats

Format	When Used	Constraints
Interactive Buttons	Binary or small choices: Confirm/Cancel, Yes/No, category selection (when 3 or fewer options)	Maximum 3 buttons per message. Each button label max 20 characters.
Numbered Text List	Lists with more than 3 items: sub-categories, services, provider results	Format: `1. Option A\n2. Option B\n3. Option C` . User replies with number.
Free Text Input	Quantities (number of birds), location (village name), dates, feedback text	Validated by conversation engine. Re-prompted on invalid input.
Media Messages	Advisory content delivery: video, audio, PDF, images	Content URL in message payload. Supported: MP4, MP3, PDF, JPEG/PNG.
Template Messages	Provider outbound notifications (order alerts, status updates)	Requires pre-approved WhatsApp Business templates. Must include all required parameters.

8. Location & Matching

Location Capture

- Farmer provides location as **free text** (village/ward/county) or **GPS live location** (WhatsApp location share)
- **Google AI normalises** free text to structured `County + Sub-County`
- System confirms normalised location with farmer before proceeding ("*Nakuru County, Naivasha Sub-County. Correct?*" [Yes] [No])

Provider Matching

- Providers register **explicit service areas**: one or more County + Sub-County combinations
- **Match logic:** farmer's `county + sub_county` must exist in provider's registered service areas
- A provider near a county border can register **neighbouring counties/sub-counties**

Kenya Geography

- **47 Counties**, each with multiple Sub-Counties
- Pilot will start with selected counties and expand

9. Payment Model

PILOT CONSTRAINT

All payments are **offline and self-reported**. There is no online payment processing in the pilot.

Aspect	Details
Accepted methods	Cash and M-Pesa only
Processing	All payments happen offline — no online payment gateway
Farmer reports	Payment method (Cash / M-Pesa) via WhatsApp buttons
Provider reports	Payment received (yes/no), amount, method via Kuza Web App
Reconciliation	No automated reconciliation in pilot. Feedback flow handles disputes.
Market flow	Payment direction is **reversed**: provider/aggregator pays farmer

10. Provider Onboarding & Catalog Management

Existing Agripreneurs

Agripreneurs already exist in the Kuza Platform `user` table. They need supplementary data added:

- Service offerings (linked to catalog items)
- Per-unit pricing
- Service areas (County + Sub-County selections)

New Provider Onboarding

Full onboarding flow for providers not already in Kuza Platform:

1. **Registration** — name, phone, provider type (vet, agrovet, aggregator, trainer, etc.)
2. **Service area setup** — select one or more Counties + Sub-Counties
3. **Catalog creation** — list services with category, sub-category, pricing, capacity
4. **Price list management** — per-unit pricing, availability windows

All management is performed through the **Kuza Web App**. Catalog is published to the Beckn network via `BPP → ONIX → Catalog Service`.

11. Integration Points

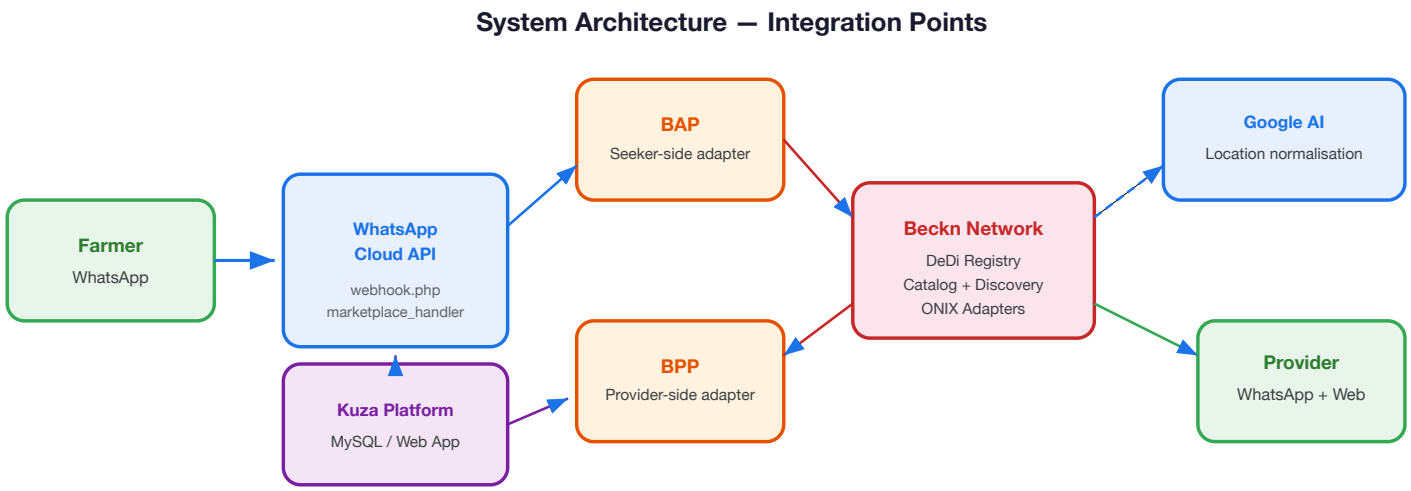

Figure 11.1 — High-level integration architecture

11.1 WhatsApp Cloud API

Direction	Mechanism	Details
Inbound	`webhook.php`	Receives all WhatsApp messages, routes marketplace flows to `beckn_marketplace_handler.php`
Outbound to farmer	`beckn_marketplace_handler.php`	Sends interactive messages, text, media via WhatsApp Cloud API
Outbound to provider	Messaging queue / cron	Triggers pre-approved template messages with action button links

11.2 Beckn Network

- **BAP** registered with DeDi Registry (identity + public key + callback URI)
- **BPP** registered with DeDi Registry
- **ONIX adapters** deployed on GCP (Docker containers)
- Catalog published to **Beckn Catalog Service**

- Discovery routed through **Beckn Discovery Service**
- Transaction messages (`select` / `init` / `confirm` / `status`) routed directly BAP ↔ BPP via DeDi Registry

11.3 Google Cloud Services

Google contributes across three distinct service tiers. Each is consumed at a different point in the pipeline.

11.3.1 Google Cloud Platform (GCP) — Hosting Infrastructure

- Hosts the entire Kuza-built stack: WhatsApp Handler, BAP, BPP, Kuza Web App, ONIX adapters, MySQL database
- Provides networking, TLS termination, load balancing, public endpoints for BAP / BPP callbacks
- Co-locates Kuza application code with Google Content APIs and AI Interfaces for low-latency in-flow calls

11.3.2 Google Content APIs — Curated Domain Content

Google publishes **curated content APIs per domain** (agri, health, market, finance, etc.) that serve *informational* / *educational* content to farmers via the chatbot. These are **separate from Beckn transactional catalog data** — they don't list providers or services for sale, they deliver knowledge.

- **Primary consumer:** Advisory flow — the BPP calls Google Content APIs during `on_discover` processing to source advisory material (articles, videos, audio, infographics, structured guides).
- **Secondary consumers:** Livelihood / Household / Market flows can optionally enrich a transaction with supplementary content (e.g., "Best practices for vaccinating broilers" attached to a Vaccination order; "Egg storage tips" attached to a Market produce sale).
- **Domain coverage:** agri, health, market, finance, education — matched to Kuza's flow categories, with each domain having its own taxonomy aligned to the master data taxonomy in Kuza Platform.
- **Delivery format:** APIs return content payloads (text, media URLs, structured guides) that the BPP wraps into the Beckn `on_discover` callback for Advisory, or surfaces as supplementary messages in the chatbot for other flows.
- **Called from:** BPP application logic; optionally from the WhatsApp Handler for inline content.

11.3.3 Google AI Interfaces — Intelligent Processing

Distinct from content APIs, the AI interfaces provide *computational* services that process input and return derived results.

- **Location normalisation (all flows):** Called from the PHP handler. Farmer types free-text location → handler sends to Google AI → AI returns structured County + Sub-County → handler sends confirmation to farmer on WhatsApp.
- **Advisory content tailoring (Advisory flow):** Called from the BPP after retrieving curated content from Content APIs. The AI personalises, summarises, or translates the source material based on farmer context (location, enterprise, language) before delivery.
- **Future:** intent extraction from free-text farmer queries, language translation (English ↔ Swahili), sentiment analysis on feedback.
- **Called from:** PHP handler (location normalisation) and BPP (content tailoring, future intent extraction).

> **Content APIs vs AI Interfaces — how they relate:** Content APIs serve *what we know* (curated knowledge bases per domain). AI Interfaces serve *what we infer* (normalising, generating, translating). The Advisory flow typically uses both: Content API supplies the source material, AI Interface tailors it for the specific farmer.

11.4 Kuza Platform

- BAP and BPP connect to **MySQL** for all data operations
- Provider catalog read from MySQL tables
- Orders, payments, feedback written to MySQL tables
- Conversation state managed in MySQL tables

12. Non-Functional Requirements

Requirement	Specification
No shopping cart	Each order is a single service/product transaction. No multi-item baskets.
No shipping	Services delivered on-site or at provider location. Market produce collected at farm gate or collection point.
No online payments	All payments are offline, self-reported by both parties.
No app installation	Farmer uses WhatsApp only. Provider uses WhatsApp + browser-based web app.
Offline-friendly	Critical WhatsApp flows work on low bandwidth. Minimal data transfer per interaction.
Multilingual	System should support English and Swahili at minimum.
Data privacy	Farmer phone numbers and personal data stored only in Kuza Platform, not transmitted beyond what Beckn protocol requires.
Availability	WhatsApp webhook must be highly available (provider = Meta). Kuza backend targets 99% uptime during pilot.
Response time	WhatsApp replies within 5 seconds for interactive flows. Provider discovery within 10 seconds.

13. Pilot Scope & Future Extensions

In Scope for Pilot

#	Feature	Status
1	All four flows (Livelihood, Household, Market, Advisory)	Required
2	Farmer-initiated discovery for all flows	Required
3	Provider onboarding and catalog management via web app	Required
4	Full order lifecycle for paid flows (Livelihood, Household, Market)	Required
5	Advisory content delivery (discover-only)	Required
6	Location-based provider matching (County + Sub-County)	Required
7	WhatsApp conversations for farmer + notifications for provider	Required
8	Kenya deployment (counties, KES, M-Pesa)	Required

Future Extensions (NOT in Pilot)

Extension	Description
Buyer-initiated discovery for Market	External buyers seeking produce via agripreneur aggregators
Online M-Pesa payment processing	Integrated payment gateway with real-time confirmation
Provider ratings & reviews aggregation	Cumulative rating system influencing provider ranking
Multi-language AI advisory	AI-powered content delivery in multiple local languages
Feedback-driven provider ranking	Algorithmic ranking based on completion rate, ratings, response time
Provider WhatsApp conversational menu	Catalog management via WhatsApp (no web app needed)

This document is the primary requirements specification. All development must conform to these specifications.